KYC document submission.

Additionally, India's diversity in local languages can be an obstacle, as explaining useability of AI applications to customers and spreading awareness at a local level may be difficult. However, if banks can emphasise on these aspects, customers are bound to use these services, and take India's banking systems, in general, to the next level.

Prepaid Electricity: Optimal Energy For Rural Rajasthan

Smart microgrid systems have shown 84 per cent reduction in overall energy usage in one of the villages in Rajasthan

To bring consistency in power availability across all corners of the country, Indian utilities have been urging urban users to reduce electricity wastage, increase renewable source utilisation and improve power usage methods. In the meanwhile, many innovative initiatives are being taken in rural regions to promote easy and cost-effective access to power. For instance, villages in Rajasthan have been facilitated with smart microgrid systems that deliver prepaid electricity schemes to its residents.

The setup

The solution was innovated and brought by Gram Power, a startup formed by two University of Berkley pass-outs. Here is a quick glance of the complete microgrid setup.

A power distribution unit is installed by Gram Power and connected to a solar-powered energy generation unit in the village. The selected village should have at least 100 residents or 20 households. A battery system is used to store energy.

Communication channels and smart meters are deployed along with the setup. While communication channels enable proper regulation of energy distribution, smart meters, deployed at customer units, maintain real-time energy usage data.

A local entrepreneur is contacted to manage the microgrid site. The entrepreneur buys bulk energy (generated through solar power) credits at wholesale price from Gram Power through a proprietary energy wallet.

In turn, the entrepreneur sells those credits to locals in a prepaid model and provides energy to them in accordance with the payment made. Consumers can purchase power at a minimum price of ₹ 10 per day, which can run lights for nine hours, and ceiling fan, television and other appliances for six hours, in one home.

Higher the purchase amount, higher is the energy availability. Monthly purchase options are available, too. Smart meters keep a tab on purchase balance and power being utilised. Each microgrid setup covers a radius of about two kilometres from the installation site.

The benefits

The solution has enabled overall villages to reduce energy expenses, erratic supplies and dependency on the main grid. It is reported that average power consumption per consumer in one of the villages where the setup was deployed reduced from 2.4 kilowatt-hour per day (kWh/day) to 0.4kWh/day. This is an 84 per cent reduction in overall energy usage.

Local entrepreneurs who manage the grids receive a 10 per cent commission on each power sale, which can earn them an additional ₹ 54,000 annually for every 300 connections.

Consumers receive consistent supply throughout the day. The pay-as-you-go model makes electricity affordable in rural areas, while enabling ease of access to power.

Real-time updates through smart meters help consumers avoid payment defaults. It enables the local entrepreneur to ensure that accurate amount of power is being delivered to consumers, in turn, minimising supply shortage as well as energy wastage.

Another major advantage delivered by smart meters is quick detection and prevention of power theft. This is otherwise a major nuisance in rural areas, which can result in up to 58 per cent loss of excess power.

Collateral benefits include reduced use of kerosene in villages as well as carbon footprint emitted from it. Renewable power generation reduces grid energy dependency, too.

Facing the challenges

Since the complete setup of the microgrid system must be done with assistance from local labour, lack of clear technical know-how may pose a challenge. Moreover, convincing the rural community as well as the *Panchayat* for assistance and cooperation may be difficult.

Proper maintenance and safeguarding of the system also require strict regulation. Component theft or damage remain a risk in these installations, which may need focussed vigilance.

The government has been providing aids in the form of subsidies and incentives to promote these projects and empower rural India. For instance, Ministry of New and Renewable Energy (MNRE) provides 30 per cent subsidy to off-grid projects.

The biggest USP of these systems is the flexible form factor and usage of renewable resources, which would yield long-lasting returns. While renewable energy production is costlier today, and installation and maintenance of the setup may be accompanied with difficulties, benefits, on the other hand, are multifold.

One of the earlier installations of this project was done in 2012 by Gram Power. It successfully led to expansion of the projects in more rural regions. Further aid by the government and active participation from more solution providers from similar backgrounds can substantially reduce the supply-demand gap of power throughout rural India. EFY